

Yani Aliche

Olathe, KS | (913)-286-9882 | yanialiche07@gmail.com | www.linkedin.com/in/yanialiche

Engineering Portfolio: <https://yanialiche.github.io/>

EDUCATION

Cornell University, Duffield College of Engineering

Ithaca, NY

B.S. in Mechanical Engineering, Intended Minor Aerospace Engineering

Aug 2025 – May 2029

Relevant Coursework: Engineering Operations: Data Science and Decision Making, Statics & Mechanics of Solids, Differential Equations for Engineers, Thermodynamics, Physics II: Electromagnetism; *Spring 2027*: Dynamics, Introduction to Mechanical Design

SKILLS

Technical: Fusion 360, SolidWorks, AutoCAD, 3D printing, Microsoft Office Suite (Excel, Word, PowerPoint)

Programming: Python, MATLAB

Language: English (Native), Tamazight (Native)

EXPERIENCE

The Home Depot

Apr 2024 – Aug 2026

- Promoted from Cashier to Delivery and Hardware Associate based on reliability and performance; advised customers on product specifications and applications while fulfilling orders and deliveries

PROJECTS & RESEARCH

Independent Research: Airfoil Surface Design and Aerodynamic Performance

Sept 2024 – May 2025

- Fabricated and adapted 5 3D-printed airfoils with varied surface patterns to evaluate aerodynamic efficiency
- Analyzed lift and drag data using experimental design methods to identify performance differences
- Presented at the Greater KC Science & Engineering Fair; awarded *1st Place, Engineering Mechanics Division*

Airfoil Design & Analysis Project | SolidWorks

Aug 2025 – Oct 2025

- Modeled research airfoils in SolidWorks, extending prior experimental work into a computational framework
- Applied CFD and FEA simulations to analyze lift-to-drag ratios and wing stress under aerodynamic loads

Open-Fan Engine Design Project | Fusion 360 & SolidWorks

Feb 2026 – Mar 2026

- Designed a conceptual open-fan jet engine (inspiration from CFM RISE program) using Fusion 360 and SolidWorks
- Applied parametric modeling, CAD surfacing, and assembly techniques across a full propulsion system concept.

Automotive Powertrain Design | SolidWorks

July 2026

- Designed and assembled a virtual automotive powertrain with a running engine, transmission, driveshaft, and differential assemblies
- Developed an understanding of how power is transferred from the engine through the transmission, driveshaft, and differentials while applying parametric modeling and mechanical mates.

ACTIVITIES & LEADERSHIP

Intramural Sports Supervisor – Cornell Recreational Services

May 2026 – Present

- Lead intramural soccer, basketball, and volleyball operations for 200+ weekly participants.
- Mentor and train new referees on rules, game control, and conflict resolution.
- Support staff leadership in resolving disputes, evaluating referees, and maintaining safety standards.

Co-President – Muslim Student Association (MSA)

Aug 2024 – May 2025

- Oversaw budgeting, fundraising, and event planning
- Organized a fundraiser with local high schools, raising \$4,000 to support cultural and educational initiatives

CREDENTIALS

- Cornell Tradition Fellow *Aug 2025 – Present*
- OIE Engineering Scholar *Aug 2025 – Present*
- GKCSEF Engineering Mechanics — Senior Division Winner *Mar 2025*